Model-Free Safe Reinforcement Learning Through Neural Barrier Certificate

Yujie Yang, Yuxuan Jiang, Yichen Liu, Jianyu Chen, and Shengbo Eben Li, *Senior Member, IEEE*

***Abstract*—Safety is a critical concern when applying reinforcement learning (RL) to real-world control tasks. However, existing safe RL works either only consider expected safety constraint violations and fail to maintain safety guarantees, or use overly conservative safety certificate tools borrowed from safe control theory, which sacrifices reward optimization and relies on analytic system models. This letter proposes a model-free safe RL algorithm that achieves near-zero constraint violations with high rewards. Our key idea is to jointly learn a policy and a neural barrier certificate under stepwise state constraint setting. The barrier certificate is learned in a model-free manner by minimizing the violations of appropriate barrier properties on transition data collected by the policy. We extend the single-step invariant property of the barrier certificate to a multi-step version and construct the corresponding multi-step invariant loss. This loss balances the bias and variance of the barrier certificate and enhances both the safety and performance of the policy. The policy is optimized under the constraint of the multi-step invariant property using the Lagrangian method. We optimize the policy in a model-free manner by introducing an importance sampling weight in the constraint. We test our algorithm on multiple problems, including classic control tasks, robot collision avoidance, and autonomous driving. Results show that our algorithm achieves near-zero constraint violations and high performance compared to the baselines. Moreover, the learned barrier certificates successfully identify the feasible regions on multiple tasks.**

***Index Terms*—Robot safety, reinforcement learning, neural barrier certificate.**

I. Introduction

REINFORCEMENT learning (RL) has achieved promising performance in many challenging control tasks, including video games [1], board games [2], robotics [3], and autonomous driving [4]. RL solves the optimal policy by maximizing the expected sum of rewards. However, in many real-world control tasks such as robotics and autonomous driving, the policy not only needs to maximize the rewards but also must maintain strict safety guarantees [5], [6], [7], [8]. Learning a policy with zero constraint violations while optimizing rewards remains a challenge.

Many safe RL algorithms deal with safety under the framework of constrained Markov decision process (CMDP) [9], which constrain the expected sum of safety violation costs below a given threshold. These algorithms achieve low costs in many safe control tasks [10], [11], [12], [13], [14], [15]. However, they cannot achieve zero constraint violations. Most of them set the cost threshold to be a positive value, which allows a small number of violations. Even if the threshold is set to zero, the constraint violations still cannot be eliminated since the cost signal becomes extremely sparse when the number of constraint violations is close to zero [16].

Another line of work use safety certificates in control theory to ensure constraint satisfaction [7], [16], [17], [18], [19], [20], [21], [22], [23]. The safety certificates, e.g., barrier certificates [22] or Lyapunov certificates [17], specify a forward invariant subset of the state space, in which the agent will never violate the safety constraints. However, the limitations of these algorithms are that they either require strong assumptions of safety certificates [7], [16], [18], [19] or rely on environment models [17], [21], [22], [23], [24]. Predefined assumptions may result in an overly restrictive safety certificate and overly conservative policy. Accurate environment models are often unavailable in complex tasks. Inaccurate models, either known a priori or learned through data, may result in sub-optimal policies.

To overcome the restrictive assumptions of safety certificates and requirements for environment models of existing safety certificate methods, in this paper, we propose a safe RL algorithm that jointly learns a policy and a neural barrier certificate in a model-free manner, as shown in Fig. 1. The barrier certificate guarantees that the agent does not leave a forward invariant feasible set so that the safety constraint is always satisfied. The barrier certificate is represented using neural networks without any predefined rules or handcrafted functional forms. The learning of both the barrier certificate and the policy do not require any prior known or learned environment models. The barrier certificate is learned by penalizing the violations of appropriate barrier properties on state samples collected by the policy. The policy is learned by solving a constrained optimization problem, which is converted into an unconstrained optimization problem using the Lagrangian method. While most existing methods require analytic environment models to back-propagate the gradient of the constraint to the policy [17], [21], [22], [23], [24], we penalize the policy using the violation of the barrier invariant

Manuscript received 10 September 2022; accepted 4 January 2023. Date of publication 20 January 2023; date of current version 27 January 2023. This letter was recommended for publication by Associate Editor P. D. Lillo and Editor J. P. Desai upon evaluation of the reviewers' comments. This work was supported in part by NSF China under Grant 52221005, in part by the Tsinghua University Initiative Scientific Research Program, and in part by Tsinghua University-Toyota Joint Research Center for AI Technology of Automated Vehicle. *(Yujie Yang and Yuxuan Jiang contributed equally to this work.) (Corresponding author: Shengbo Eben Li.)*

Yujie Yang, Yuxuan Jiang, and Shengbo Eben Li are with the School of Vehicle and Mobility, Tsinghua University, Beijing 100084, China (e-mail: yangyj21@mails.tsinghua.edu.cn; jyx21@mails.tsinghua.edu.cn; lisb04@gmail.com).

Yichen Liu is with the Institute of Interdisciplinary Information Sciences, Tsinghua University, Beijing 100084, China (e-mail: liuyichen0518@gmail.com).

Jianyu Chen is with the Institute of Interdisciplinary Information Sciences, Tsinghua University, Beijing 100084, China, and also with Shanghai Qizhi Institute, Shanghai 200232, China (e-mail: jianyuchen@tsinghua.edu.cn).

The source code can be found at https://github.com/jjyyxx/srlnbc.

Digital Object Identifier 10.1109/LRA.2023.3238656

2377-3766 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

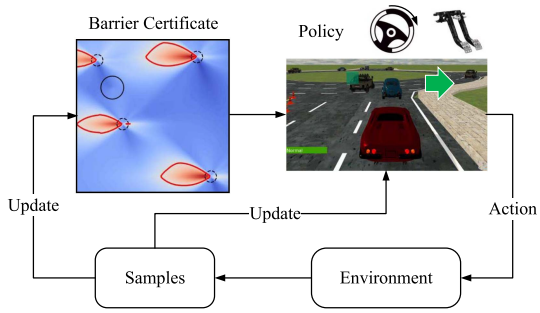


Fig. 1. Model-free safe RL through neural barrier certificate. The barrier certificate distinguishes infeasible regions (red) from feasible regions (blue) in the state space. It penalizes the actions that enter infeasible regions or approach them too quickly. The policy is learned by maximizing accumulative rewards on data samples while constrained by the properties of the barrier certificate.

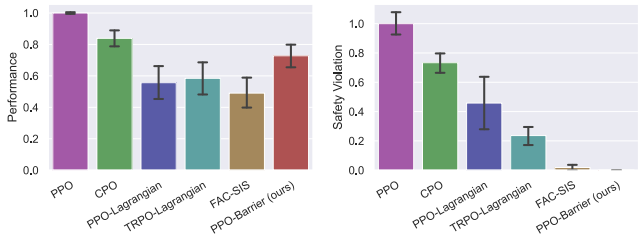


Fig. 2. Normalized average performance and safety violation on three Safety Gym environments, i.e., point goal, car goal, and doggo goal. The scores of all algorithms are normalized by that of PPO. The performance is the average episode return after convergence over three runs on each task. The safety violation is the average number of constraint violations in an episode after convergence.

property on state samples and thus do not require an environment model.

Our main contributions can be summarized as follows:

1) We propose a model-free safe RL algorithm that jointly learns a policy and a neural barrier certificate to guarantee constraint satisfaction. Our algorithm requires neither a predefined barrier certificate nor any environment models.We learn both the policy and the neural barrier certificate in a model-free manner. The barrier certificate is learned by minimizing the violations of its properties on transition data collected by the policy. We multiply the violations by importance sampling factors to construct policy constraints, which do not require an analytic system model.
2) We extend the single-step invariant property of the barrier certificate to a multi-step version and constructs a corresponding multi-step invariant loss to facilitate the learning of the barrier certificate. The proposed loss balances the bias and variance of the neural barrier certificate and enhances both safety and performance of the policy.
3) We evaluate our algorithm in multiple environments, including classic control tasks, robot collision avoidance, and autonomous driving. Results show that our algorithm achieves near-zero constraint violations as well as comparable or higher performance than the baselines. It also successfully learns the feasible regions on multiple tasks. Fig. 2 shows example statistical results on Safety Gym.

II. Related Work

A. Constrained Markov Decision Process (CMDP)

Algorithms based on CMDP constrains the expected discounted accumulative costs below a given threshold. They approximate the accumulative costs by a cost value function, which is learned in a similar way to the value function in RL. Most of these algorithms use the Lagrangian method to convert the original constrained problem into an unconstrained problem and solve it using dual ascent [10], [25]. Tessler et al. [12] incorporates the constraint as a penalty signal into the reward function. Ding et al. [26] combines primal-dual method with natural policy gradient. In order to dampen the oscillation of dual ascent, Stooke et al. [15] uses PID control to update the Lagrange multiplier. Another method is to use convex approximations of the objective function and constraint in a trust region. Achiam et al. [11] proposes Constrained Policy Optimization (CPO), which augments TRPO [27] with constraints and uses trust-region optimization to compute the analytic solution of the policy. Other algorithms use a two-step projection method. Yang et al. [13] first performs a reward improvement update and then projects the policy back onto the constraint set. Zhang et al. [14] first finds the optimal policy in the non-parametric space and then projects it back into the parametric space. The drawback of CMDP-based algorithms is that they cannot achieve zero constraint violations. This is because the cost signal becomes extremely sparse as the number of constraint violations is close to zero, which makes it difficult to learn the cost value function [16].

B. Safe Control Through Safety Certificates

Safe control algorithms consider the setting in which safety is specified through state constraints. They constrain the policy update using safety certificates, which guarantee that the policy will stay in a feasible region in the state space. However, these algorithms have limitations in either of the following two ways. Some of them require handcrafted safety certificates [7], [19], [20], while in complex control tasks, designing a good safety certificate requires a lot of trial and error. Ma et al. [16] addresses this limitation by learning the parameters of the safety certificate together with the policy, but the functional form of the certificate is still assumed a priori. Others rely on analytic environment models to compute the safety certificate and the policy [17], [21], [24], [28], while in real-world control tasks, accurate environment models are often unavailable. Lindemann et al. [29] learns hybrid CBFs for hybrid systems with known dynamics to synthesize safe control inputs. Robey et al. [30] learns CBFs with provable safety guarantees in known nonlinear control affine dynamical systems with safe expert trajectories. Lafmejani et al. [31] relies on labeled data to pretrain a CBF and use it as a constraint for model predictive control. Several works address this limitation by considering uncertain models or learning models from data. Brunke et al. [32] learns the CBF condition using Bayesian linear regression, and provides safety guarantees during learning in an uncertain system. Wang et al. [33] uses Gaussian processes to learn the partially unknown dynamics of the system and expands the barrier certified safe region based on an adaptive sampling scheme. Qin et al. [34] learns barrier certificates for black-box dynamical systems with an inaccurate nominal model. Luo and Ma [22] learns a stochastic environment model through data and uses adversarial training

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

to ensure robust safety. However, learning environment models brings extra computational burdens, and model errors may result in sub-optimal policies.

III. Preliminaries

A. Problem Statement

We study the problem of RL under state constraints. We consider discrete-time deterministic environment dynamics,

$$x_{t+1} = f(x_t, u_t), \tag{1}$$

where $x \in X$ is the state, $u \in U$ is the control input and f is the transition function. The initial state is randomly sampled from a initial set $X_{\text{init}} \subseteq X$. Safety is specified through state constraints: $h(x_t) \leq 0,\ \forall t \geq 0$, where $h : X \to \mathbb{R}$ is the constraint function. We define the safe set as $X_{\text{safe}} = \{x|h(x) \leq 0\}$ and the unsafe set as $X_{\text{unsafe}} = \{x|h(x) > 0\}$. We assume that $X_{\text{init}} \subseteq X_{\text{safe}}$. Safety only specifies the constraint in a single step. To account for long-term constraint satisfaction, we define the concept of feasibility, also known as recusive feasibility in the control community.

Definition 1: (Feasibility). If a state x and a policy π satisfy the following conditions:

$$\forall t \geq 0, x_t \in X_{\text{safe}}, u_t = \pi(x_t), x_0 = x. \tag{2}$$

Then, x is a (endlessly) feasible state under policy π. If all states in X_{init} are feasible under policy π, π is a feasible policy.

To distinguish Definition 1 from initial feasibility in finite horizon problems, we also call it endless feasibility, and use the subscript "Edls"[35]. We denote the set of all feasible policies as Π_{Edls}. Our objective is to find a policy π that is feasible and maximizes the expected discounted sum of rewards,

$$\arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(x_t, u_t)\right],\ \text{s.t. } \pi \in \Pi_{\text{Edls}}. \tag{3}$$

where γ is the discount factor and r is the reward function. We denote the solution to (3) as π^*.

B. Barrier Certificate

Barrier certificate of a certain policy is a function of the state that specifies a feasible regionforward invariant set in the state space. and keeps the agent in this regionThe control input computed by the policy keeps the system safe in this region.

Definition 2: (Barrier certificate) $B^\pi : X \mapsto \mathbb{R}$ is a barrier certificate of policy π if it satisfies the following properties:

1) $\forall x \in X_{\text{init}}, B^\pi(x) \leq 0$.
2) $\forall x \in X_{\text{unsafe}}, B^\pi(x) > 0$.
3) $\forall x \in X,\ B^\pi(x') - B^\pi(x) \leq -\alpha(B^\pi(x))$, where $x' = f(x, \pi(x))$ and α is a strictly increasing function with $\alpha(0) = 0$ (i.e. an extended class-$\mathcal{K}$ function).

A common choice for $\alpha(B^\pi(x))$ is simply λB^π for some $0 < \lambda < 1$, which is also adopted in this paper. An intuitive explanation for property 3) is that the barrier certificate should not increase too fast below zero and should decrease fast enough above zerothe rate of change over time of the barrier certificate should not be too large. A barrier certificate specifies a forward invariant set in the state space, which is its sub-level set $X_B^\pi = \{x|B^\pi(x) \leq 0\}$, i.e., for every state x in X_B^π, $x' = f(x, \pi(x))$ is still in X_B^π. It is proven that if the initial state $x_0 \in X_B^\pi$, the agent will never enter the unsafe set X_{unsafe} under policy π [36], [37]. Since forward invariance is mainly guaranteed by property 3), we call this property the invariant property. The concept of forward invariance and feasibility are closely connected. Specifically, all the states in X_B^π are feasible under policy π.

IV. Learning Barrier Certificate

In this section, we discuss our method for learning a neural barrier certificate. We design the loss functions based on the barrier properties.The barrier certificate is updated by minimizing a properly designed loss function on sampled data. We propose a multi-step invariant loss to balance the bias and variance of the barrier certificate.

A. Data Collection

We design our algorithm under the framework of reinforcement learning, where data is collected through environment exploration. The data set, which we denote as $\mathcal{D}$, contains states, actions, and rewards at each time step. We further define two subsets of $\mathcal{D}$, $\mathcal{D}_{\text{init}}$ and $\mathcal{D}_{\text{unsafe}}$. $\mathcal{D}_{\text{init}}$ is the set of initial state samples, defined as $\mathcal{D} \cap X_{\text{init}}$, thus assumed to be safe. $\mathcal{D}_{\text{unsafe}}$ is the set of unsafe state samples, defined as $\mathcal{D} \cap X_{\text{unsafe}}$.

B. Loss Function

We approximate the barrier certificate using neural networks with parameters ϕ and denote it as $B_\phi^\pi(x)$. The loss function of the barrier certificate is designed to penalize the violations of its properties in Definition 2. It consists of three terms which are the feasible loss, the infeasible loss, and the invariant loss. The feasible loss corresponds to property 1),

$$J_{\text{Edls}}(\phi) = \frac{1}{|\mathcal{D}_{\text{init}}|} \sum_{x \in \mathcal{D}_{\text{init}}} \max(B_\phi^\pi(x), 0). \tag{4}$$

where $\mathcal{D}_{\text{init}}$ is the set of state samples in the initial set. The infeasible loss function corresponds to property 2),

$$J_{\text{Infs}}(\phi) = \frac{1}{|\mathcal{D}_{\text{unsafe}}|} \sum_{x \in \mathcal{D}_{\text{unsafe}}} \max(-B_\phi^\pi(x), 0). \tag{5}$$

where $\mathcal{D}_{\text{unsafe}}$ is the set of state samples in the unsafe set. The invariant loss function corresponds to property 3),

$$J_{\text{Invt}}(\phi) = \frac{1}{|\mathcal{D}|} \sum_{x, x' \in \mathcal{D}} \max(B_\phi^\pi(x') - (1-\lambda)B_\phi^\pi(x), 0). \tag{6}$$

where $\mathcal{D}$ is the set of all state samples. The barrier certificate loss is the sum of the three terms,

$$J_B(\phi) = J_{\text{Edls}}(\phi) + J_{\text{Infs}}(\phi) + J_{\text{Invt}}(\phi). \tag{7}$$

We update the parameters ϕ by minimizing (7) using gradient descent.

C. Multi-Step Invariant Property

The invariant property of the barrier certificate only specifies the relationship between values of neighboring states. In fact, this property can be applied iteratively through time steps,

$$B_\phi^\pi(x_{t+n}) \leq (1-\lambda)B_\phi^\pi(x_{t+n-1}) \leq \cdots \leq (1-\lambda)^n B_\phi^\pi(x_t), \tag{8}$$

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

which formulates an n-step invariant property. Accordingly, the invariant loss (6) can be extended to an n-step version,

$$J^n_{\text{Invt}}(\phi) = \frac{1}{|\mathcal{D}|}\sum_{x_t\in\mathcal{D}} \max(B^\pi_\phi(x_{t+n}) - (1-\lambda)^n B^\pi_\phi(x_t), 0). \tag{9}$$

In theory, minimizing the one-step invariant loss 6(6) is enough to guarantee the invariant property of the barrier certificate. In practice, however, the barrier certificate is approximated using neural networks and the loss function is computed based on a bootstrap mechanism which uses the estimated barrier certificate to update itself. This is similar to temporal difference (TD) in RL. At an early stage of the training process, the barrier certificate is inaccurate and provides poor signals for update. If we only use one-step invariant loss, the learning process will be unstable and slow to converge. Multi-step invariant loss reduces the effect of biased barrier estimates by multiplying $B^\pi(x)$ by a smaller coefficient, $(1-\lambda)^n$. Meanwhile, this leads to a larger variance because the trajectory is randomly sampled from the environment and a longer trajectory has larger variance. This is similar to the effect of n-step TD, which has smaller bias and larger variance than one-step TD. To balance bias and variance, we take a weighted average of (9) with different steps, which is similar to TD(λ).,

$$J^\gamma_{\text{Invt}}(\phi) = \sum_{n=1}^{N} \frac{\gamma^{n-1} J^n_{\text{Invt}}(\phi)}{\sum_{m=1}^{N}\gamma^{m-1}}, \tag{10}$$

where N is the maximum number of steps we consider. The weights of future time steps are decayed exponentially by a factor of γ ($0<\gamma<1$). Our experiments show that the multi-step invariant loss 10(10) stabilizes and accelerates the learning process.

V. Joint Learning of Barrier Certificate and Safe Policy

Using the multi-step invariant loss (10), we obtain the loss function of the barrier certificate,

$$J_B(\phi) = J_{\text{Edls}}(\phi) + J_{\text{Infs}}(\phi) + J^\gamma_{\text{Invt}}(\phi). \tag{11}$$

The policy is learned by solving a constrained optimization problem, in which the expected discounted sum of rewards is maximized under the constraint of the barrier certificate. Any RL algorithm can be used to maximize the rewards. In this paper, we choose proximal policy optimization (PPO) [38]. The objective function of the policy,

$$J_\pi(\theta) = \frac{1}{|\mathcal{D}|}\sum_{x,u\in\mathcal{D}} \min\left(\rho(\theta)A^{\pi_{\text{old}}}(x,u),\right.$$
$$\left.\text{clip}\left(\rho(\theta), 1-\epsilon, 1+\epsilon\right)A^{\pi_{\text{old}}}(x,u)\right), \tag{12}$$

where θ is the parameter of the policy network, $\rho(\theta) = \frac{\pi_\theta(u|x)}{\pi_{\text{old}}(u|x)}$ is the importance sampling factor, and $A^{\pi_{\text{old}}}$ is the advantage function computed using generalized advantage estimation [39]. The constraint of the policy is the invariant property of the barrier certificate, which guarantees that the agent will never leave the feasible region. We use the multi-step invariant loss to construct the policy constraint. The policy optimization problem is,

$$\max_\theta \; J_\pi(\theta)$$
$$\text{s.t. } \rho(\theta)J^\gamma_{\text{Invt}}(\phi) \le 0. \tag{13}$$

In the constraint of (13), we use importance sampling because the data used for computing the invariant loss is collected by an old policy π_{old}. We convert (13) into an unconstrained problem using the Lagrangian method,

$$\min_{\nu\ge 0}\max_\theta L(\nu,\theta) = J_\pi(\theta) - \nu\rho(\theta)J^\gamma_{\text{Invt}}(\phi). \tag{14}$$

We solve (14) using dual ascent method, in which θ and ν are iteratively updated using gradient descent. The value function is learned by minimizing the mean square error,

$$J_V(w) = \frac{1}{|\mathcal{D}|}\sum_{x,u\in\mathcal{D}}\left(V_w(x) - (A^{\pi_{\text{old}}}(x,u) + V_{\bar{w}}(x))\right)^2, \tag{15}$$

where w and $\bar{w}$ are the parameters of the value and target value networksand $\hat{R}(x)$ is the estimated return. The pseudocode of our algorithm is in Algorithm 1.

Algorithm 1. Safe RL through Barrier Certificate

Initialize: policy network parameters θ, value network parameters w, barrier certificate parameters ϕ, Lagrange multiplier ν, learning rate α.
for *each iteration* **do**
  Initialize sample set $\mathcal{D}\leftarrow\emptyset$ and initial state x_0.
  for *each evironment step* t **do**
    Observe the state x_t.
    Compute the control input: $u_t = \pi_\theta(x_t)$.
    Apply u_t and receive the reward r_t.
    Store the samples: $\mathcal{D}\leftarrow\mathcal{D}\cup\{x_t,u_t,r_t\}$.
  end
  Update the Lagrange multiplier:
  $\nu\leftarrow\nu-\alpha\nabla_\nu L(\nu,\theta)$.
  for *each gradient step* **do**
    Update the policy network:
    $\theta\leftarrow\theta+\alpha\nabla_\theta L(\nu,\theta)$.
    Update the value network:
    $w\leftarrow w-\alpha\nabla_w J_V(w)$.
    Update the barrier certificate:
    $\phi\leftarrow\phi-\alpha\nabla_\phi J_B(\phi)$.
  end
end

VI. Experiments

A. Classic Control Tasks

We test our algorithm on two classic control tasks to examine whether the learned feasible regions of the barrier certificates match the ground truth. The environment dynamics of these two tasks are known so that we can compute the analytic feasible regions of the optimal policy.

The first task is emergency braking control, in which the agent should stop a car before it hits the wall. The state consists of the distance to the wall and the velocity of the car. The control input is the acceleration of the car. The reward is the decrement of

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

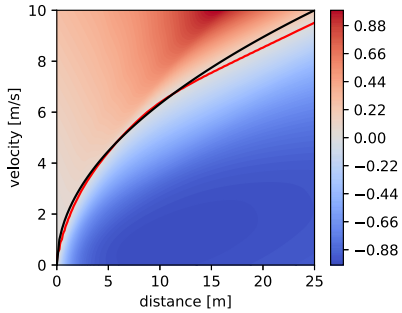
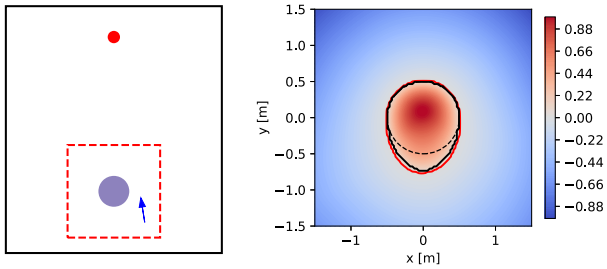
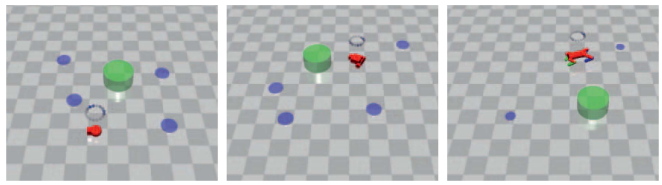


Fig. 3. Learned barrier certificate of emergency braking control. The black curve is the ground-truth boundary of the feasible region. The red curve is the zero contour of the learned barrier certificate.



(a) Reach-avoid control (b) Learned barrier certificate

Fig. 4. The blue circle is the hazard and the red circle is the goal. The black dashed line is the hazard and the black solid line is the ground-truth boundary of the feasible region. The ground-truth is obtained by simulation on a dense grid employing a maximum steering and deceleration policy. The red line is the zero contour of the learned barrier certificate.



(a) Point goal (b) Car goal (c) Doggo goal

Fig. 5. Safety Gym environments. The green cylinders are the goals and the blue circles are hazards.

TABLE I
HYPER-PARAMETERS

Algorithm	Hyper-parameter	Value
Shared	Number of hidden layers	2
	Hidden layer size	256
	Activation function	Tanh
	Discount factor	0.99
	Learning rate for value and policy	3e-4
PPO	Batch size	20000
	GAE parameter	0.97
	Clip parameter ϵ	0.2
PPO-Lagrangian TRPO-Lagrangian CPO	Initial Lagrange multiplier	1.0
	Learning rate for Lagrange multiplier	1e-2
	Backtrack coefficient	0.8
	Backtrack iterations	10
FAC-SIS	Batch Size	64
	Replay Buffer Size	500000
	SIS interval	24
PPO-Barrier	Learning rate for barrier	3e-4
	Barrier parameter λ	0.1
	Number of barrier steps N	20

distance to the wall, which encourages the agent to drive as close to the wall as possible. The task has a max time step of 200. The learned barrier certificate is shown in Fig. 3. It shows that the learned feasible region is close to the ground truth. The error is a little larger in the upper right corner because the states in this area are less frequently sampled.

The second task is reach-avoid control, in which the agent should control a point robot to reach a goal while avoiding a hazard, as shown in Fig. 4 (left). The state consists of the position, velocity, and yaw of the robot. The control input is the acceleration and yaw rate. The reward is the decrement of distance to the goal, which encourages the agent to go as fast as possible towards the goal. The task has a max time step of 100. The robot is randomly initiated inside the red dashed line, and we plot the learned barrier certificate in this region, as shown in Fig. 4 (right). It shows that the shape of the learned feasible region matches the ground truth, and the size is a little smaller. This conservativeness makes the agent less likely to enter the infeasible region.

B. Safety Gym

We test our algorithm on the safe RL benchmark environment Safety Gym [40]. In our experiments, the task is to reach a goal while avoiding hazards. The robots include point, car, and doggoWe choose three environments, point goal, car goal, and doggo goal, in which the task is to control different robots to reach a goal while avoiding hazards, as shown in Fig. 5. The observation includes the robot's state, the goal position, and lidar points of the hazards. The state includes the robot's sensor readings, the relative position of the goal, and lidar points of the hazards. In point goal and car goal, the robot's sensor readings include acceleration, velocity, angular velocity, and heading angle. In doggo goal, apart from the above sensor readings, the positions and velocities of the robot's joints and the contact forces with the ground are also included. The action of point goal includes torques of the moving and turning motors. The action of car goal includes torques of the left and right wheel motors. The action of doggo goal includes torques of 12 motors for controlling the joints. The reward consists of the decrement of distance to the goal and a bonus of reaching the goal. The cost is +1 when the robot enters the hazards and zero otherwise. Each task has a max time step of 1000.

The baseline algorithms for comparison include PPO-Lagrangian [40], TRPO-Lagrangian [40], CPO [11], FAC-SIS [16], and PPO [38]. The first three algorithms are designed under the CMDP setting. We set their cost thresholds to zero since our aim is to achieve zero constraint violation. FAC-SIS jointly learns a policy and a safety certificate with a predefined functional form. PPO is included to show the performance and constraint violations of an unconstrained algorithm. Detailed hyper-parameters are in Table I.

We evaluate the episode cost and episode return of the algorithms, as shown in Fig. 6. In all three environments, our algorithm achieves near-zero constraint violations and high returns

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

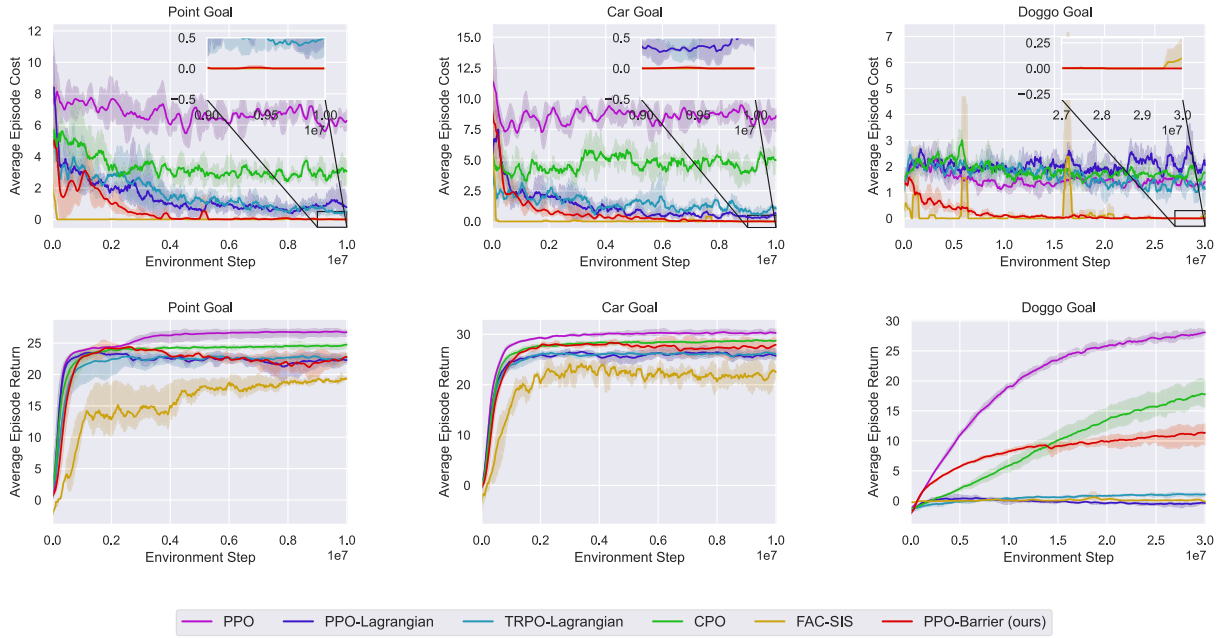
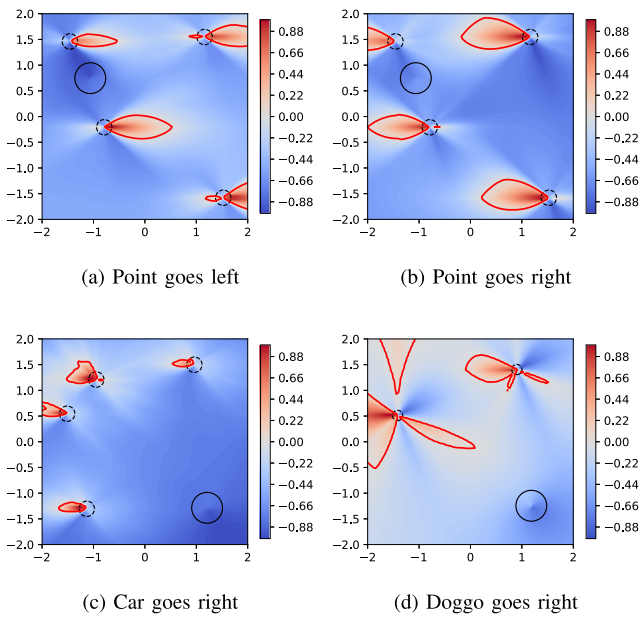


Fig. 6. Average episode cost and return on Safety Gym. The solid lines correspond to the mean and the shaded regions correspond to 95% confidence interval over three runs.

compared to the baselines. The three CMDP-based algorithms keep violating the constraints, and their costs are as high as that of PPO in doggo goal. FAC-SIS also achieves near-zero constraint violations in point goal and car goal. However, its cost curve is not stable and has several high peaks in doggo goal, which indicates that its safety may degrade in complex control tasks. In addition, the return performance of FAC-SIS is lower than our algorithm in all three environments, especially in doggo goal, which indicates that the predefined safety certificate is too conservative.

We plot the learned barrier certificates on three Safety Gym environments in Fig. 7. The solid black circle is the goal, and the dashed black circles are the hazards.The input of the barrier certificate is generated by fixing the velocity and orientation of the robot, placing it on a grid, and computing the state at each grid point. The result shows that the infeasible regions locate next to the hazards on the same side where the robot approaches them. This indicates that the barrier certificate not only learns the locations of the hazards but also learns the orientations and dynamics of the robot. On doggo goal, the learned infeasible regions look irregular and not as good as those on the other environments. The main reason is that it is difficult to set proper values for positions and velocities of the joints and the contact forces with the ground. We simply set them to zeros, which may mismatch the state distribution during training.

We do ablation studies on the number of barrier steps. Results are shown in Fig. 8. The result shows that a step of one, which is adopted by most existing algorithms [7], [18], [21], [22], is not able to ensure safety. When the multi-step barrier is used, a step of five or above can ensure near-zero violation. This indicates that our proposed multi-step invariant loss is effective to ensure safety. Moreover, as the barrier step increases, the return performance also increases. This is because the multi-step invariant loss balances the bias and variance of the barrier certificate, which stabilizes and accelerates the training process. Generally speaking, a larger number of steps is preferred. This is because a small number of steps will result in high bias. However, a larger number of steps also increases the computation burden because all the n-step losses need to be computed in each gradient step. In our experiments, we find that a step of 20 is enough to perform well on all environments, including complex ones like doggo goal and MetaDrive (see Section VI-C).



(a) Point goes left (b) Point goes right

(c) Car goes right (d) Doggo goes right

Fig. 7. Barrier certificates on Safety Gym. The speed of the robots are 1 m/s. The solid black circle is the goal, the dashed black circles are the hazards, and the red lines are zero-level contours of the barrier certificates.

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

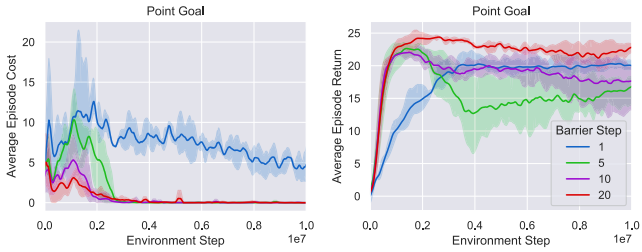


Fig. 8. Ablation studies on the number of barrier steps on point goal.

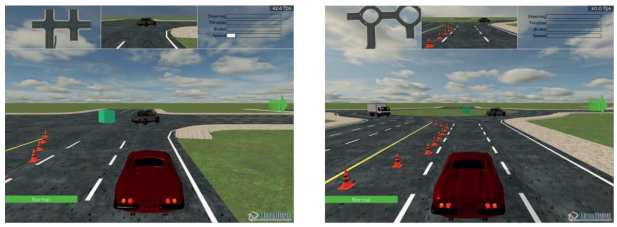


Fig. 9. Examples of MetaDrive traffic scenarios used in experiments.

C. Metadrive

We test our algorithm on MetaDrive [41], a compositional driving simulator that can generate an unlimited number of traffic scenarios, as shown in Fig. 9. The topology of the road network is randomly composed from elementary components, e.g., straight road, curve road, intersection, and roundabout. Each road network has a start point and a destination, connected with a route in the form of a series of navigation points. In our experiment, the environment contains 10 different road networks. The task is to drive the ego vehicle to a given destination while avoiding collisions with surrounding vehicles and other traffic objects, including traffic cones, barriers, and warning triangles. MetaDrive has realistic vehicle dynamics with chassis, wheels and joints simulated by Panda3D Physics Engine, which is equivalent to a 14 DOF full vehicle model. The observation includes the state of the ego vehicle, the navigation points, and the lidar points of surrounding objects. The control input includes the steering angle and acceleration. The reward function contains a dense driving reward for longitudinal movement towards the navigation point and a sparse terminal reward for arriving at the destination. The cost is +1 when the ego vehicle collides with surrounding objects and zero otherwise. The task terminates when the ego vehicle arrives at the destination or drives off the road, or reaches a max time step of 1000.

We choose PPO-Lagrangian, TRPO-Lagrangian, and CPO as baseline algorithms for comparison. Detailed hyper-parameters are in Table I. The evaluation metrics are average success rate and average episode cost, which are also adopted by the original MetaDrive paper [41]. Success rate is the ratio of episodes in which the agent arrives at the destination. Episode return is not used because success rate is a more suitable measurement in this environment, as explained in [41]. We plot the learning curves in Fig. 10. Results show that our algorithm achieves near-zero constraint violations while the costs of the baselines remain high. Moreover, our algorithm achieves a success rate of over 0.8, comparable to that of TRPO-Lagrangian and much higher than that of CPO and PPO-Lagrangian.

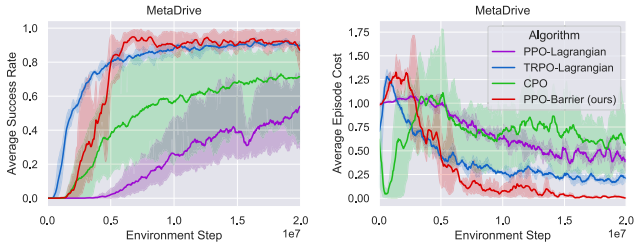


Fig. 10. Average success rate and episode cost on MetaDrive.

VII. Limitation

While our proposed algorithm achieves near-zero safety violations on high dimensional complex control tasks, solid safety is not guaranteed. Due to approximation error of neural networks, the learned barrier certificate may fail to guarantee forward invariance on some states. Moreover, we suppose that the training is carried out in simulation environments, where safety violations do not cause real-world damage. Thus, we only focus on reducing safety violations of the policy at convergence. During the learning phase, there will likely be safety constraint violations. Deploying the policy to real world requires consideration of model uncertainty and robust safety, which we will address in our future work.

VIII. Conclusion

In this letter, we propose a model-free safe RL algorithm that jointly learns a policy and a neural barrier certificate to guarantee constraint satisfaction while optimizing accumulative rewards. The barrier certificate is learned by minimizing the violations of barrier properties computed on state samples. We propose a multi-step invariant loss function to facilitate the learning of the barrier certificate, which balances the bias and variance of the barrier network. The policy is optimized under the constraints of the multi-step barrier invariant property certificate using the Lagrangian method. Experimental results on classic control tasks, robot collision avoidance, and autonomous driving show that our algorithm achieves near-zero constraint violations as well as comparable or higher performance than the baselines. The barrier certificate also successfully learns the feasible regions on multiple tasks.

References

[1] L. Kaiser et al., "Model-based reinforcement learning for atari," in *Proc. Int. Conf. Learn. Representations*, 2020.
[2] J. Schrittwieser et al., "Mastering atari, go, chess and shogi by planning with a learned model," *Nature*, vol. 588, no. 7839, pp. 604–609, 2020. [Online]. Available: https://doi.org/10.1038/s41586-020-03051-4
[3] M. Andrychowicz et al., "Learning dexterous in-hand manipulation," *Int. J. Robot. Res.*, vol. 39, no. 1, pp. 3–20, 2020.
[4] B. R. Kiran et al., "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022.
[5] J. Garcıa and F. Fernández, "A comprehensive survey on safe reinforcement learning," *J. Mach. Learn. Res.*, vol. 16, no. 1, pp. 1437–1480, 2015.
[6] Y. Chow, O. Nachum, E. Duenez-Guzman, and M. Ghavamzadeh, "A Lyapunov-based approach to safe reinforcement learning," in *Proc. Neural Inf. Process. Syst.*, 2018, pp. 8103–8112.
[7] R. Cheng, G. Orosz, R. M. Murray, and J. W. Burdick, "End-to-end safe reinforcement learning through barrier functions for safety-critical continuous control tasks," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, pp. 3387–3395.

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.

[8] L. Brunke et al., "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annu. Rev. Control, Robot., Auton. Syst.*, vol. 5, pp. 411–444, 2021.
[9] E. Altman, *Constrained Markov Decision Processes*. Boca Raton, FL, USA: CRC, 1999, vol. 7.
[10] Y. Chow, M. Ghavamzadeh, L. Janson, and M. Pavone, "Risk-constrained reinforcement learning with percentile risk criteria," *J. Mach. Learn. Res.*, vol. 18, no. 1, pp. 6070–6120, 2017.
[11] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 22–31.
[12] C. Tessler, D. J. Mankowitz, and S. Mannor, "Reward constrained policy optimization," in *Proc. Int. Conf. Learn. Representations*, 2019.
[13] T.-Y. Yang, J. Rosca, K. Narasimhan, and P. J. Ramadge, "Projection-based constrained policy optimization," in *Proc. Int. Conf. Learn. Representations*, 2020.
[14] Y. Zhang, Q. Vuong, and K. Ross, "First order constrained optimization in policy space," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 15338–15349.
[15] A. Stooke, J. Achiam, and P. Abbeel, "Responsive safety in reinforcement learning by PID lagrangian methods," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 9133–9143.
[16] H. Ma, C. Liu, S. E. Li, S. Zheng, and J. Chen, "Joint synthesis of safety certificate and safe control policy using constrained reinforcement learning," in *Proc. 4th Annu. Learn. Dyn. Control Conf.*, 2021, vol. 168, pp. 97–109.
[17] Y.-C. Chang, N. Roohi, and S. Gao, "Neural Lyapunov control," in *Proc. Adv. neural Inf. Process. Syst.*, 2019, vol. 32, pp. 3245–3254.
[18] A. Taylor, A. Singletary, Y. Yue, and A. Ames, "Learning for safety-critical control with control barrier functions," in *Proc. Learn. Dyn. Control.*2020, pp. 708–717.
[19] W. Zhao, T. He, and C. Liu, "Model-free safe control for zero-violation reinforcement learning," in *Proc. Conf. Robot Learn.*, 2021, pp. 784–793.
[20] P. L. Donti, M. Roderick, M. Fazlyab, and J. Z. Kolter, "Enforcing robust control guarantees within neural network policies," in *Proc. Int. Conf. Learn. Representations*, 2021.
[21] Z. Qin, K. Zhang, Y. Chen, J. Chen, and C. Fan, "Learning safe multi-agent control with decentralized neural barrier certificates," in *Proc. Int. Conf. Learn. Representations*, 2021.
[22] Y. Luo and T. Ma, "Learning barrier certificates: Towards safe reinforcement learning with zero training-time violations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, vol. 34, pp. 25621–25632.
[23] C. Dawson, Z. Qin, S. Gao, and C. Fan, "Safe nonlinear control using robust neural Lyapunov-barrier functions," in *Proc. Conf. Robot Learn.*, 2021, pp. 1724–1735.
[24] Y.-C. Chang and S. Gao, "Stabilizing neural control using self-learned almost Lyapunov critics," in *Proc. Int. Conf. Robot. Automat.*, 2021, pp. 1803–1809.
[25] Y. Chow, A. Tamar, S. Mannor, and M. Pavone, "Risk-sensitive and robust decision-making: A cvar optimization approach," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1522–1530.
[26] D. Ding, K. Zhang, T. Basar, and M. Jovanovic, "Natural policy gradient primal-dual method for constrained markov decision processes," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, vol. 33, pp. 8378–8390.
[27] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1889–1897.
[28] S. M. Richards, F. Berkenkamp, and A. Krause, "The Lyapunov neural network: Adaptive stability certification for safe learning of dynamical systems," in *Proc. 2nd Conf. Robot Learn.*, B. Aude, D. Anca, P. Jan, and M. Jun, Eds., 2018, vol. 87, pp. 466–476.
[29] L. Lindemann et al., "Learning hybrid control barrier functions from data," in *Proc. Conf. Robot Learn.*, 2021, pp. 1351–1370.
[30] A. Robey et al., "Learning control barrier functions from expert demonstrations," in *Proc. IEEE 59th Conf. Decis. Control*, 2020, pp. 3717–3724.
[31] A. Salimi Lafmejani, S. Berman, and G. Fainekos, "NMPC-LBF: Nonlinear MPC with learned barrier function for decentralized safe navigation of multiple robots in unknown environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2022, pp. 10297–10303.
[32] L. Brunke, S. Zhou, and A. P. Schoellig, "Barrier bayesian linear regression: Online learning of control barrier conditions for safety-critical control of uncertain systems," in *Proc. Learn. Dyn. Control Conf.*, 2022, pp. 881–892.
[33] L. Wang, E. A. Theodorou, and M. Egerstedt, "Safe learning of quadrotor dynamics using barrier certificates," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2018, pp. 2460–2465.
[34] Z. Qin, D. Sun, and C. Fan, "Sablas: Learning safe control for black-box dynamical systems," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 1928–1935, Apr. 2022.
[35] S. E. Li, *Reinforcement Learning for Sequential Decision and Optimal Control*. Berlin, Germany: Springer, 2023.
[36] A. D. Ames, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs with application to adaptive cruise control," in *Proc. IEEE 53rd Conf. Decis. and Control.*, 2014, pp. 6271–6278.
[37] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *Proc. 18th Eur. Control Conf.*, 2019, pp. 3420–3431.
[38] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:.06347*.
[39] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *Proc. Int. Conf. Learn. Representations*, 2016.
[40] A. Ray, J. Achiam, and D. Amodei, "Benchmarking safe exploration in deep reinforcement learning," 2019, *arXiv:1910.01708*.
[41] Q. Li, Z. Peng, Z. Xue, Q. Zhang, and B. Zhou, "Metadrive: Composing diverse driving scenarios for generalizable reinforcement learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2022.

Authorized licensed use limited to: Tsinghua University. Downloaded on January 28,2023 at 07:27:15 UTC from IEEE Xplore. Restrictions apply.